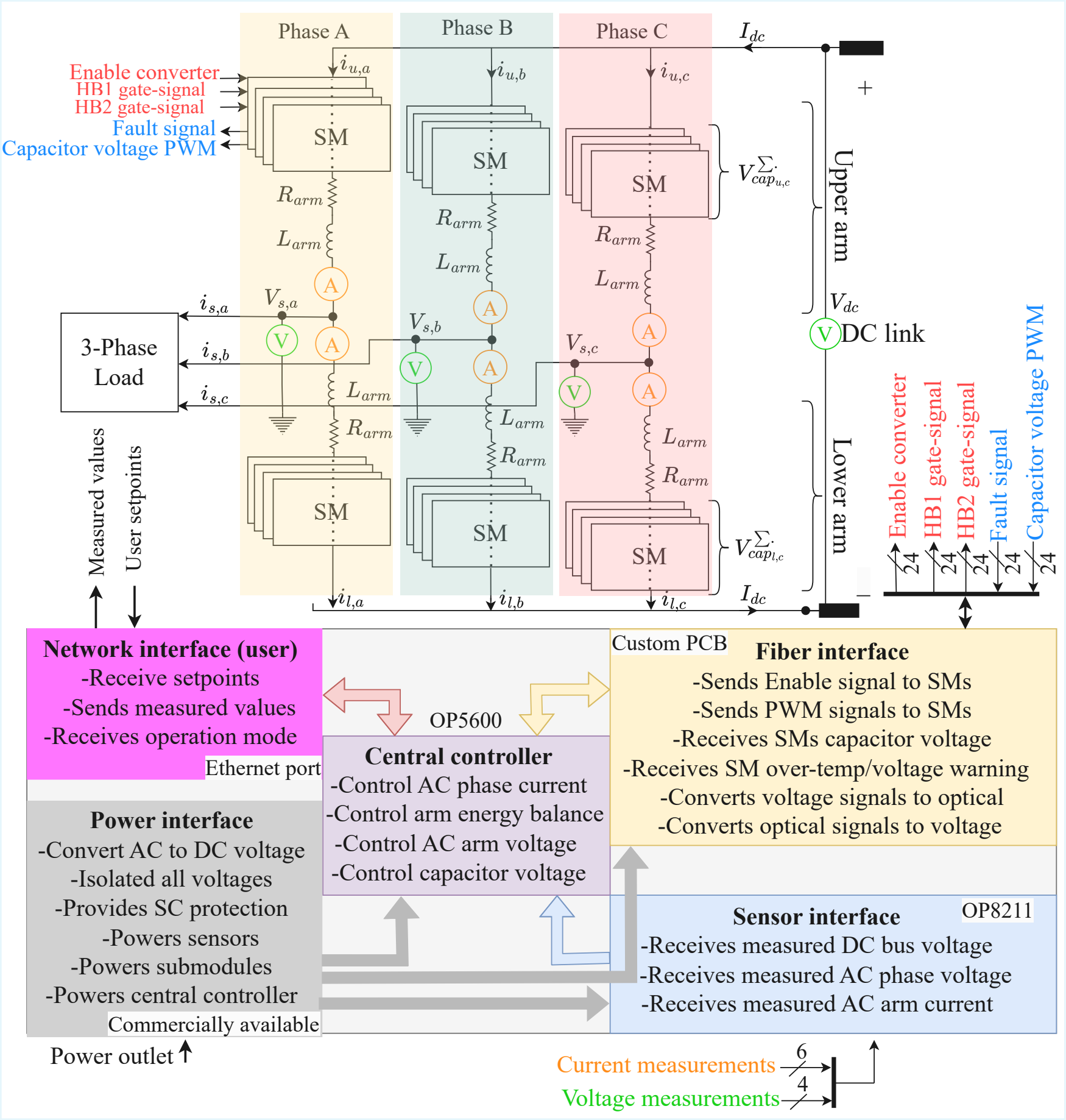
Phase A
Phase B
Phase C
I_{dc}
$i_{u,a}$
$i_{u,b}$
$i_{u,c}$
Enable converter
HB1 gate-signal
HB2 gate-signal
Fault signal
Capacitor voltage PWM
SM
R_{arm}
L_{arm}
$V_{cap_{u,c}}^{\Sigma}$
Upper arm
V_{dc}
DC link
$V_{s,a}$
$V_{s,b}$
$V_{s,c}$
$i_{s,a}$
$i_{s,b}$
$i_{s,c}$
3-Phase Load
$V_{cap_{l,c}}^{\Sigma}$
Lower arm
$i_{l,a}$
$i_{l,b}$
$i_{l,c}$
Measured values
User setpoints
24
Network interface (user)
-Receive setpoints
-Sends measured values
-Receives operation mode
Ethernet port
Custom PCB
Fiber interface
-Sends Enable signal to SMs
-Sends PWM signals to SMs
-Receives SMs capacitor voltage
-Receives SM over-temp/voltage warning
-Converts voltage signals to optical
-Converts optical signals to voltage
OP5600
Central controller
-Control AC phase current
-Control arm energy balance
-Control AC arm voltage
-Control capacitor voltage
Power interface
-Convert AC to DC voltage
-Isolated all voltages
-Provides SC protection
-Powers sensors
-Powers submodules
-Powers central controller
Commercially available
Sensor interface
OP8211
-Receives measured DC bus voltage
-Receives measured AC phase voltage
-Receives measured AC arm current
Power outlet
Current measurements
Voltage measurements
6
4